

Qatar-1b

R-1385

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_191025 · created 2026-08-01T02:20:23+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458781.6244 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.056
Transit depth (Rp/Rs)^2	3.66 +/- 2.16 [%]
Orbital Inclination (inc)	88.45 +/- 2.85
Airmass coefficient 1 (a1)	3.24 +/- 1.36
Airmass coefficient 2 (a2)	-0.99 +/- 0.82
Scatter in the residuals of the lightcurve fit is	38.46 %
Best Comparison Star	None
Optimal Aperture	1.82
Optimal Annulus	3.0
Transit Duration (day)	0.0827 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.191 $\pm$ 0.056 vs 0.1463 (deviation 0.8 σ)
Duration fit vs published	0.0827 d vs 0.0692 d (deviation 1.93 σ)
Mid-time O-C	-3.7 min
Depth SNR	3.4
Residual scatter	38.46 %

Light curve

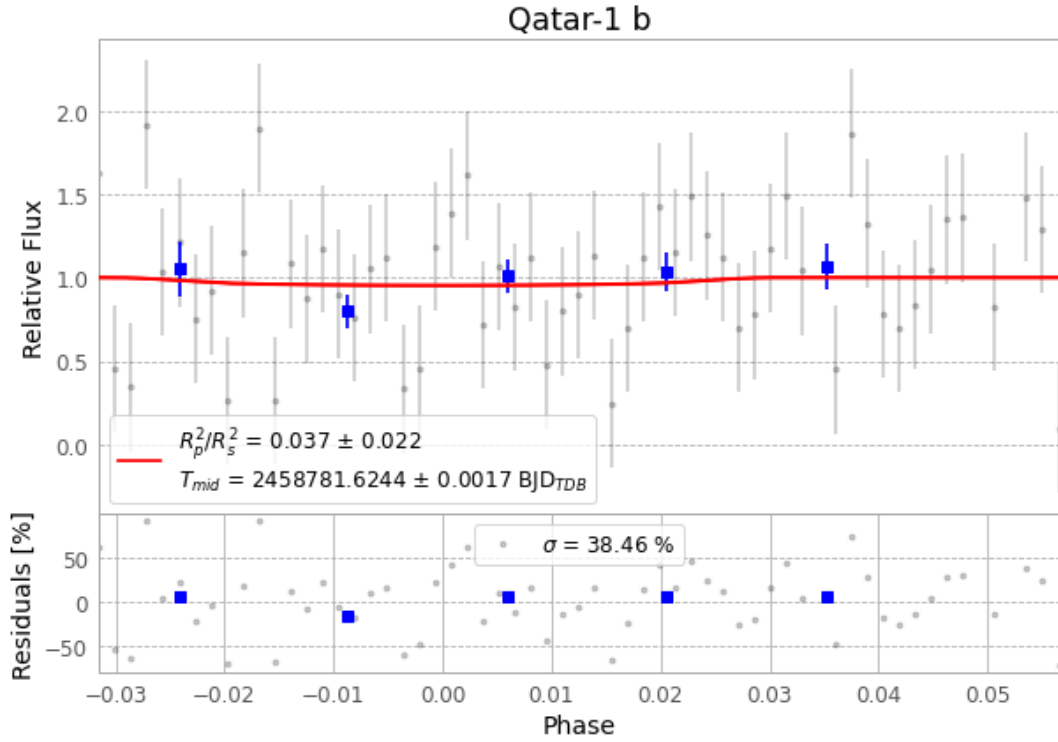


Figure 1 — FinalLightCurve_Qatar-1 b_2019-10-24.png (SHA-256 6050d8679d7c4257...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [161, 316], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 f1294015d0856863...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

61 FITS frames (40 MB), Qatar-1191025015112.FITS → Qatar-1191025045112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-191025.log (8fd747d4c5ae...), FinalLightCurve_Qatar-1 b_2019-10-24.png (6050d8679d7c...), FinalLightCurve_Qatar-1 b_2019-10-24.csv (da330a8b0554...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)